

Experiment 10.2

Z-Transform and Inverse Z-Transform

Program (Scilab)

(Requires Symbolic Toolbox)

```
// =====  
// Numerical Z-Transform  
// =====  
// Given sequence (example)  
x = [1, 0.5, 0.25, 0.125]; // length L  
N = 64; // Number of points for DFT  
n = 0:length(x)-1;  
// Define z-points on unit circle (frequency domain)  
k = 0:N-1;  
omega = 2 * %pi * k / N;  
z = exp(%i * omega); // unit circle points  
// Compute Z-transform on unit circle (DTFT)  
Xz = zeros(1, N);  
for m = 1:N  
Xz(m) = sum(x .* (z(m) .^ (-n)));  
end  
// Plot Magnitude and Phase  
clf;  
subplot(2,1,1);  
plot(omega/(2*%pi), abs(Xz));  
xtitle("Magnitude Spectrum", "Normalized Frequency", "|X(z)|");  
subplot(2,1,2);
```

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plot(omega/(2*%pi), atan(imag(Xz)./real(Xz)));
xlabel("Phase Spectrum", "Normalized Frequency", "Phase (rad)");

// =====

// Numerical Inverse Z-Transform

// =====

// We use inverse DFT to recover x[n]

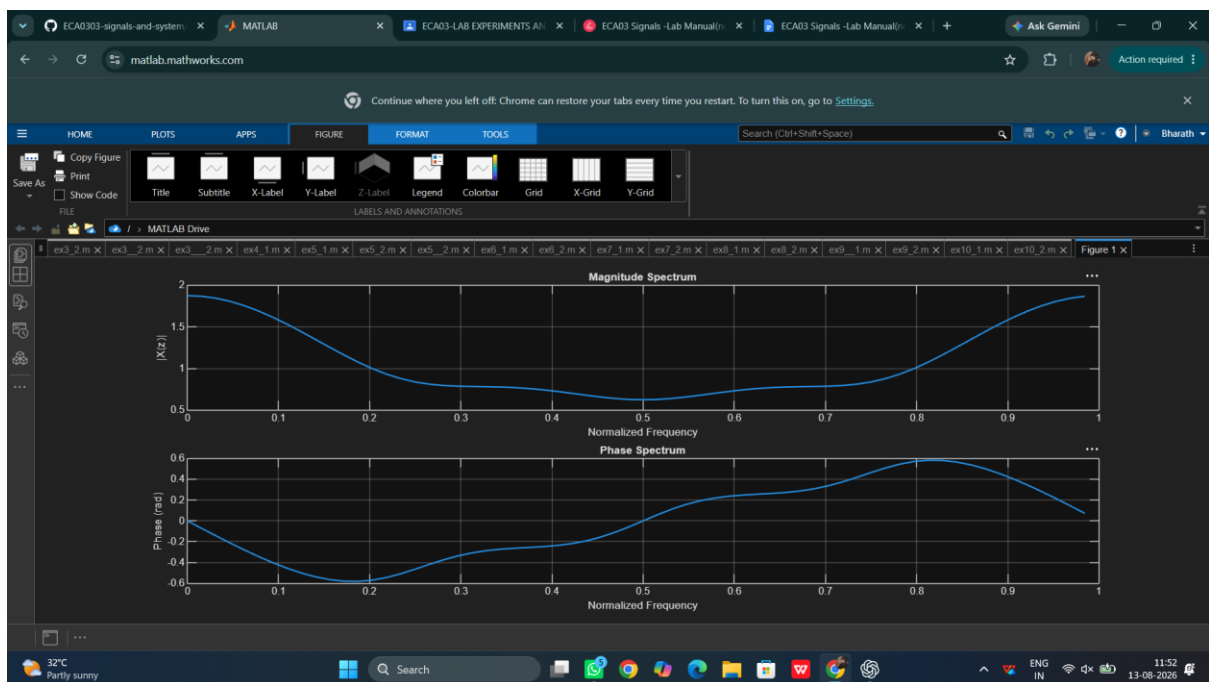
x_rec = zeros(1, N);

for m = 1:N
x_rec(m) = (1/N) * sum(Xz .* (z.^ (n(m)))));
end

disp("Recovered sequence from Inverse Z-Transform:");
disp(real(x_rec(1:length(x))));

```

Output Generated Graph



Result

Z-transform derived for given sequence.